MUHAMMAD NAJWAN SULVADLI

+62 812 9337 8651 | sulvadliwork@gmail.com | LinkedIn Profile | Portfolio

A versatile Technology Engineer and Systems Architect specializing in the intersection of Artificial Intelligence, IoT, and Data Engineering. With a strong foundation in both hardware integration and full-stack software development, I have a proven ability to transform complex operational challenges into streamlined, automated solutions. I bring a unique blend of academic excellence and hands-on experience in deploying robust, scalable technologies for high-stakes industries such as mining, construction, and telecommunications.

Work Experiences

PT Global Service Indonesia (Contracted to PT Kalimantan Prima Persada) – Jakarta, Indonesia

End to End AI Engineer & Data Analyst *Feb 2026 – Present*

- Automated Operational Monitoring: Developed and deployed an AI-based vehicle load monitoring system using live CCTV feeds to replace manual visual inspections, reducing dependency on human oversight by approximately 80% through 24/7 automated surveillance.
- Enhanced Safety Compliance: Engineered an AI safety monitoring system for workshop environments to automatically detect Personal Protective Equipment (PPE) compliance, improving safety monitoring efficiency by over 70% through automated violation reporting.
- Digital Attendance Transformation: Launched an AI-driven facial recognition system for employee attendance, eliminating 100% of manual attendance sheets and reducing HR administrative processing time by ~90%.
- Advanced Incident Investigation: Implemented a multi-camera zone intrusion system that automatically captures safety violations, significantly cutting post-incident investigation time compared to manual CCTV review.
- Internal Automated Document Review Plagiarism: Architected a modular plagiarism automated and optical character recognition, replacing manual detection platform that automates document review processes, enabling simultaneous processing of multiple files and reducing manual effort.

IoT Engineer May 2025 – Jan 2025

- Integrated Communication Solutions: Solo-designed and deployed an end-to-end Push-to-Talk (PTT) dispatch system for mining operations, improving audio communication efficiency by ~80% across multiple sites.
- Infrastructure Optimization: Developed a specialized communication client and relay server for ruggedized mining hardware, minimizing communication downtime and reducing IT setup time for new field devices..
- Operational Visibility: Built a web-based operator console with real-time status tracking, providing the control room with a single unified interface to manage all field communications and unit movements.
- Hardware & Firmware Integration: Developed custom firmware for ESP32-based devices and implemented reliable data pipelines to ensure stable data flow from the field to the cloud.

Antares – PT Telkom Indonesia – Jakarta, Indonesia Dec 2024 – Mar 2025

AI & Embedded Systems Developer (Internship)

- Developed AI vision and machine learning solutions for edge computing, optimizing real-time performance.
- Built multiple Computer Vision AI models optimized for real-time inference, enhancing project efficiency at Antares.
- Successfully implemented Nvidia Jetson and Raspberry Pi Hailo AI kit for AI vision-based edge computing solutions, improving accuracy and processing capabilities.
- Enhanced AI model performance, contributing to higher-quality AI solutions and increasing Antares technological capabilities.

PT TAS - Vendor for PT Adhi Karya – Jakarta, Indonesia Jan 2024 – Jan 2025

Programmer Freelance / Project-Based

- Designed and developed automation plugins for cost calculations and safety equipment visualization in Autodesk Revit, supporting PT Adhi Karya's construction projects.
- Improved efficiency of BIM Modeller and QHSE teams by up to 80% through technical solutions based on visual programming.
- Integrated visual programming to simplify cost calculation and safety equipment visualization processes.
- Successfully implemented DBMS in PT Adhi Karya's admin dashboard web application to manage safety equipment price data effectively and efficiently.

Honeywell Connected Laboratory UI – Depok, Indonesia Jun 2023 – Aug 2023

Laboratory assistant / Assistant lecturer

- Developed automated control systems using Programmable Logic Controller (PLC) to enhance laboratory performance efficiency.
- Integrated Internet of Things (IoT) in industrial equipment maintenance processes to monitor equipment health in real-time.
- Successfully implemented a web-based system integrated with IoT for visual industrial equipment maintenance.
- Managed and configured Linux-based systems to support laboratory operations.

Education Level

Asia e University (AeU), School of Science & Technology – Selangor, Malaysia Sep 2021 – Sep 2025

Bachelor Degree in Information Communication and technology: ***CGPA: 3.62 / 4.00***

- Developed object-oriented programming skills in Java and C++, focusing on software design and development.
- Completed academic projects, including an IoT-based final project integrated with a mobile web app.
- Gained knowledge of IT project management, business processes, and system analysis.

University of Indonesia, CEP-CCIT Faculty of Engineering – Depok, Indonesia Sep 2021 – Sep 2023

Associate Degree in Information Technology – Internet Based System Automation: ***CGPA: 3.83 / 4.00***

- Acquired experience in Internet of Things (IoT) technology, including sensor networks, embedded systems, and cloud computing.
- Configured and designed networking devices, including routers and switches using Cisco Packet Tracer.
- Learned Android application development using Java and Kotlin in Android Studio.
- Developed websites integrated with Database Management Systems (DBMS) using frameworks like Laravel and CodeIgniter.

Achievement:

- Best Graduate in TI – Internet-based System Automation, 2024.

Professional Experience

Research Club Mentor – Mentari Intercurtural School Jakarta – Jakarta, Indonesia Sep 2024 – Dec 2024

- Guided students in developing innovative projects in the Internet of Things (IoT), software engineering and Android application development.
- Provided technical training and foundational concepts in programming and IoT-based application development and mobile development.
- Supported students in research, design, and implementation of technology-based projects.
- Directed students in presenting their projects at school and national-level technology events and competitions.

Assistant Lecturer – Office Staff Training, University of Indonesia – Depok, Indonesia Jul 2023 – Aug 2023

- Successfully assisted 30 participants from the administrative staff of the University of Indonesia.
- Supported lecturers in helping participants learn Microsoft Office tools, including Word, Excel, and PowerPoint.

Programming Team – Autonomous Unmanned Aerial Vehicle (AUAV) UI – Depok, Indonesia Jan 2023 – Jan 2024

- Member of the programming team at AUAV, Universitas Indonesia, working on fixed-wing aircraft.
- Successfully implemented AI-based object recognition systems for visual-based applications.

Student Contract – Cisco Networking Academy – Depok, Indonesia Sep 2022 – Des 2022

- Participated in all training activities at the Cisco Networking Academy, Faculty of Engineering UI.
- Studied IoT Fundamentals: Big Data & Analytics, IoT Security & Connecting Things.

Assistant Lecturer – Badiklat IoT Pusdiklat Tekfunghan Kemhan – Jakarta, Indonesia Oct 2022 – Nov 2022

- Successfully assisted 43 participants in IoT device training with ESP, Arduino, Microcontrollers, Actuators, Sensors, and various IoT-based applications (Blynk, Thingspeak, Thinger.io).
- Supported lecturers in guiding participants in learning IoT and programming.

Assistant Lecturer – Python training for Unsurya students – Depok, Indonesia Aug 2022 – Sep 2022

- Successfully assisted 30 participants from Suryadarma University.
- Guided participants in learning Python programming using PyCharm.

Skills

- **Technical skills:** Docker, Ultralytics, HuggingFace, Azure Devops, Databricks, Power Bi, Prompting, Internet of Things (IoT), Arduino, Virtual machine, Linux, Android, Object Oriented Programming (OOP), Programming Languages (Java, Python, C++, Kotlin), Google common product, Corel Draw, Vegas, Cisco Packet Tracer.
- **Soft skills:** Teamwork collaboration, Problem solving, Time management and organization, Adaptability and flexibility, Leadership and management and Active listening.
- **Language:** Bahasa Indonesia (native), English (fluent).